

ActInf Livestream #007.0: "Variational ecology and the physics of sentient

Livestream | 1:20:16

all right

hello and welcome everyone this is the

active inference live stream

i am daniel friedman and this is the

active inference live stream

double o 7.0 welcome

to team com team com is an experiment in

online team communication

learning and practice related to active

inference

you can find us on twitter at

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via our public key based team or on

youtube

this is a recorded and archived live

stream

please provide us feedback so that we

can improve our work all backgrounds and

perspectives are welcome here

video etiquette for live streams is that

you want to mute

if there's any sound in your background

raise your hand if you want to

speak so you can be added to the stack

use respectful speech etc

today in act imp stream 7.0

i'm going to try to set the context for

7.1 and 7.2

which is going to be a discussion of the

paper variational ecology and the

physics of sentient systems by ramstad

at

all 2019 in physics of life reviews

this video as the other dot zeros are is
just an introduction
to some context it's not a review or a
final word
the goal is to provide context for the
ideas
philosophical threads and vocabulary of
that ramstead at all
paper go through some other points as
well and the punch line
is that ecosystems are multi-scale
systems with implications for how we
study
ecological organization and
the sections of 7.0 will be as follows
first we'll go through the keywords and
talk about some of the publication
context
go through the goals of the paper the
abstract of the paper and the roadmap of
the sections
go through some key quotations and then
talk a little bit about
ergotic systems some critiques of the
free energy principle
the simulations and historiosity and how
that influences
how we think about ecosystems and also
the cog ergo sum
then in 7.1 and 7.2 we're going to
discuss this paper with whoever wants to
participate so you can also save and
submit your questions
or get in touch with us if you want to
participate
also and thanks for the suggestion
to uh feedback was to how to relate this
to previous discussions

so if you want previous discussions to
listen to that relate to this
conversation

one would be active stream two is the
free energy principle a formal theory of
semantics because this
discussion and paper was asking can we
use the fep
to study meaningful systems so both the
ones that matter for us
the meaningful systems for us but then
also the meaning making the autopoietic
systems

another discussion was 5.0 and the 5.1
and 2 discussions

which are the context for this question
about internalism and externalism
which is a question about how and where
action and perception occur
are they internal to an organism or do
they

occur in this space that's mediated
between the external environment and the
internal environment

and then also this most recent 6.0
and following discussions where we
talked about tale of two densities
active inferences and active inference
because there we talked about how
inactive or ecological approaches are
broadly linked with computational
approaches like does one have precedence
over the other

all right so there's a lot going on in
this variational

ecology paper so

we're going to start here with the
keywords so

there will be pages for each of these
but here's the keywords that were
on the paper niche construction so niche
construction
is when some system of interest like an
organism
over developmental or ecological or
evolutionary time scales
modifies its environments it's pretty
broad and we'll talk more about it
in a bit variational ecology
they're using variational in a sense
here to mean just broadly multi-scale
and then akin to the processes of
variational
bayesian methods which are kind of like
general multi-scale methods
so they're thinking about ecology from a
multi-scale
perspective with a sort of flexible
computationalist
aspect as we've seen from those previous
discussions
evolutionary systems theory is uh
basically just evolutionary biology
probably some people prefer
revolutionary biology or evolutionary
systems
theory not exactly sure they're
pretty equivalent physics to the mind
will have a slide
to discuss because there's a few
different senses of what physics of the
mind entails
free energy principle of course one of
the key words here because that's
sort of the big framework or the
paradigm that everything is being

worked on or responded to under
and then variational neurology we'll
talk about more
soon but variational again here is just
meaning multi-scale
ethology no it's not ethereum it's
actually behavior
and neuro just meaning the
neurophysiological component so this is
kind of like neuro behavior
and thinking about the the cause and
effect between the
neurophysiological parameters whether
the neural or hormonal
and the behavioral outcomes and the
collective behavioral outcomes
all right so let's turn to ecology first
here's a quote from the paper they wrote
in summary the niche transcribes
regularities that pertain to group
behavior they are transcribed in the
physical layout of the niche
through the active and perceptual
dynamics of the agent
which coincides with perceptual and
active dynamics of the niche
the same dynamics entail the landscape
of action possibilities which are
affordances
which maintain the structural integrity
of the agent knees system
niche dynamics resolve the free energy
gradients that are induced by the
physical structures
of organisms in their niche and their
history of dense interaction and dynamic
coupling
in this sense the robustness of patterns

of shared intentionality and an action
of shared meaning is inherited from the
robustness of niche
and vice versa so to kind of ground this
in some other more
non-fep ecology here's a 2014
paper by deborah gordon
and here's two figures that illustrate
this point that i'll make which is
that there's going to be sort of this
hand in glove relationship
between the dynamics of the ecosystem
and the resulting collective behavioral
and quote individual behavioral
algorithms that exist
and first off the variational approach
or just the multi-scale approach
means that it's kind of collective
behavior all the way down like inside of
that ant on the bottom right
there's also a collective of neurons or
you know it's
just going to be that type of a
interactions of different scales all the
way down that's sort of the variational
realization is that it's all going to
have to be at least coarse grained
at the high in the low end if not tried
to
uh be modeled explicitly but
we can at least draw a border of
analysis
around an ant and track it as
something that we can identify as a
functional component of the system
and so in the two figures there's
two examples of axes of ecosystem
variability

so on the right side figure one it's patchiness in space and time so in a where the resource is distributed really uh evenly across space then there wouldn't be any type of mass recruitment pheromone because there'd be no reason to bring an ant to where you just got that one little blip whereas under ecological regime b it would make sense to have something like a recruitment regime because it could be an exploitable resource that might be patchy in space and time uh and you might see ants that could switch different strategies however one ant species may specialize on one or the other in figure two it's an example of a high versus low operating cost environment so when operating costs are high things are tending to be off until they're stimulated on like a fever because operating costs of a fever are high whereas when operating costs are low like for example the maintenance of leaf cutter ants under certain conditions in the rain forest to maintain territory versus the bioenergetics of moving let's just say under one example that uh is often brought up here until there's a disruption there's a continuation of function so this is sort of a similar realization

to what is described in the paper that
there's this

statistical covariation between the
collective behavioral algorithms
and the niche regularity here is another
paper by professor gordon in 2011
called diffusion of behavioral ecology
and

ecology um so on the left are just a
bunch of animals a bunch of things that
people have studied and then ethology
again is just kind of a funny
word that is used by some behavioral
biologists but not all but this is kind
of an

example of the different range of things
that people have worked on
in that area specifically and

one cool quote from this paper was until
someone discovers a fossil record
for behavior we can in practice only
investigate

what are the ecological consequences of
a trait now that is the evolutionary
ecology of behavior

so we can sometimes see for example
this you know an amber that traps uh one
insect

capturing another one we can say okay
this pickup move may have happened one
time or it may have been frequent

but all we have from a behavioral
observational perspective are actually
the live realizations of today and so
the implications are that because the
local rules of engagement and ecology
can change

it appears that selection shifts in

space and time so we know that selection pressures can go in one direction or have a certain intensity on one trait and then change in a different year or in different time

this is why behavioral ecologists who study the ecology of a trait in one time and place will find the outcomes different than another

the main project of evolutionary ecology is to understand when and why that happens

the effects of local conditions on species interactions create different ecological pressures in different places this produces the spatial mosaic of diversity and then another

quote there was context dependence requires us to redefine the project of relating behavior and fitness

it is not realistic to hope to find any reason why any trait would always be adaptive

any trait behavioral or otherwise is deeply linked to others genetically developmentally and functionally and if so of no trait so no trait has ecological impact independently of all others

moreover even if we can ascertain what variation in certain behavior is associated with variation and reproductive success

changing conditions will still shift around that relation so that's um non-stable or non-stationary fitness

landscapes

adaptive landscapes are full of small
bumps and the behavior itself modifies
the landscape

if we watch long and carefully enough we
will see different consequences for
reproductive success of the same
behavior

the project then is to learn how
conditions determine the ways that
behavior contributes to variation in
reproductive

success so behavioral ecology

already there's the realization that
it's important to think about this
interplay between features of the
organism as a

ethological unit neurobehavioral unit
but

also at the ecological the population
scale and the evolutionary scale

all right a little bit more specific
about niche construction

drawing a figure from this paper by
kendall adult

niche construction is a pretty broad
term and a lot of

outcomes and processes can be placed
under this

framework it has to do with the way that
organisms modify

their niche and this is almost

uh always going to be the case when you
have

con specifics in individuals of the same
species because

almost by definition they're going to be
taking up an overlapping

you know set of the canopy uh elevation
in a rain forest or foraging preference
and so this diagram shows how
you can have natural selection operating
to shape based on instantaneous relative
fitness

the distribution of the strategies or
genotypes and phenotypes on the right
this great line coming down here
that's genetic inheritance but then also
you have the

outcomes of collective behavior which
are influenced by

the changes in the genomic and
epigenomic

and niche and developmental factors and
cultural factors

then influencing the way that natural
selection shapes

that and that can be through semantic
information or through physical
resources like

uh eusocial insect colony so

one example here of a cool niche
construction

this is a clearing in a rain forest
and the clearing comes from these ants
that trim that live within a specific
plant and trim the ground around it a
series of paper

the devil to pay a cost of mutualism
with uh

mira militista schumanniai ants in
devil's gardens

in increased periphery *android hirsuta*
trees by frederick gordon so like
this paper is saying there's ecological
trade-offs

so it's just it's representing a
different strategy and there's
obviously a reason why these devils
gardens don't just
proliferate and sweep across the whole
area they
clearly have limiting factors so it
makes sense to study
the bigger context and understand what
these limiting factors are
and where they break down rather than
just say well it's just simply a good
strategy so it's always going to be
about kind of a bigger level of analysis
as well
all right kind of broadening from
ecology
to ecological psychology this
is related to some
often human-centric obviously as
psychology
might tend to be um
[Music]

modeling of systems around people
and we were all children and so there's
a developmental aspect
there's also these immediate to indirect
aspects of our system so there's a
multi-scale and a multi-time
aspect to ecological systems theory and
also
on the more direct psychological side
there's this three-way diagram
that comes from a 1976 paper
and there's it's sort of like a cycle
you have the on the starting on the
bottom left
the schema is what directs action and

what action accomplishes is exploration
what exploration does is it samples what
does it sample
it samples percepts like oculomotor
sampling and stimuli what does that do
it modifies your schema what would be
identified in the free energy as the
internal model
and ecological psychology also
is where we get the development of ideas
like affordances and the fields of
affordances the skilled intentionality
framework
what the organism or system can do and
that can reflect
technology as well so in our constructed
niche that can be technological
embedding
and this image is nice because it shows
the clarity of the phone the hyper
reality of the phone
in relationship to the blurriness of the
embodied
and so uh it captures that what's at
hand
what we can act on what we're using for
information
that can overwhelm our field of
performances
term that i definitely had to look up
and think about a few ways
that it applied was this physics of the
mind
i was just like okay what is that gonna
mean
and i think that there's a few different
ways i'm i'm happy if someone has a
another way to add or if there's a more

specific way to state it but
one sense of physics of the mind is that
there's a search for
lawful rules and simple equations for
the mind
to make neurosense in some sense more
like some people's conception of physics
so that there would be a loss of motion
or an invariance principle or symmetry
principle
or a relationship with some other
uh physics equation for the mind
another branch of physics for the mind
is that
whether or not it is what the brain
really does
that this branch of engineering
oriented analysis of neurosystems would
facilitate the development of systems
engineering and complex systems
and control systems so kind of like
applied physics
being engineering that's sort of where
these uh
micro sensors and brain computing
interfaces
are at it's at the implied physics level
and then another sense of physics of the
mind
is that there might be some
understanding of the physical basis
slash substrate of the mind which is
kind of the hard question of
consciousness
and it also relates to questions like
soft matter physics and crystals
quantum microtubules what what kind of
quote computation is being done in the

brain what is the brain doing
these are all kinds of questions that uh
is you know is it an
antenna or a processor is there is it
which metaphors and which aspects of
them are instructive in that regard
and the big question i'm sorry that all
of these
senses of physics the mind are getting
at is can
because the mind is explicitly a
metaphysical thing as opposed to the
brain
is can metaphysics join with multi-scale
systems neuroscience physics molecular
genetics
evolutionary ecology etc
then they mentioned you know in the
title
that it's the variational uh
ecology physics of sentient systems so
what are
sentient systems because initially i
think i had the wrong
reading here so vne in the vanc which
we're going to get to in a second
are about enabling the application of
the fep to phenomena within
and beyond the brain these approaches
hold the promise of extending the
variational
free energy approach to the dynamics of
sentient systems i.e
systems with sensory states across
spatial and temporal scales
so this is not a paper or a discussion
on consciousness or awareness this is
going to be about emergence

in multi-scale biosystems which is
a pretty compatibilist with either
interpretation
but this sentient it doesn't mean
awareness
it's not about tearing tests it's about
sense
systems so this is about being able to
model simpler
sensing systems you know thermometers
and that type of control system
in a very simple way but also more
nuanced systems like
other organisms in a more rich way
all right so how do we get here what is
the literature
context for this variational ecology
paper
the fep has recently been leveraged to
furnish a fully generalizable meta
theory
for adaptive behavior in sentient
systems across spatial and temporal
scales called variational
neuroethology so let's talk about
what is vene because this paper is going
to reference it a lot
what is vene and what scales or types of
phenomena does it model
so we're going to get to vene variational
ecology
via vene variational neuroethology
by contextualizing within a critique
that leads to this paper
so in phys life review 2018
there's a paper by ramp citadel
answering short english question which
is what is life

1944 a free life formulation
now that paper led to a bunch of
responses
which led to the clarification
on some of these responses and sort of
the drawing out
of this variational neuroethological
approach and one quote
that arises from the paper the
schrodinger's paper
was given the success of this
explanatory framework in biology we
suggest that tim bergen's levels of
inquiry might be apt to elucidate
structural
laws that supplement the general
principles provided by the fep
the fep describes a general biological
modeling imperative walton virgin has
offered a distinctive but complementary
framework that allows us to develop
substantiative
explanations for the phenotypic traits
and behaviors of any given species or
organism
and so here is the 1963 paper by tim
burgin and then a
50 years update by bateson and lot
and we talked about this in one of the
previous
discussions but this is a two by two
grid that represents tim bergens
for quote wise it's kind of like
aristotle's four why's
it's about having multiple different
kinds of explanation
that are at different types and
different scales of analysis

so the historical how how did this organism get here is development and how did it get here in the very short term the contemporary how is like the ultimate mechanism how did my arm get here because it moved from this other place ultimately or more a little bit more historically but approximately it got here because it was an arm bud and then the contemporary why is the function like what does the thing do right now just at this moment like why is there a mouse on my desk so i can use it and then historically why like why is there a mouse on my desk why is it as opposed to not using a desk or not using a mouse because of evolutionary processes so that's a little bit of a broader level question so that kind of compatibilist among uh compatibilism amongst answers is going to be complemented by a way to specifically connect multiple levels through potentially even more formal or uh modeled or testable ways of thinking about the relationship between these four categories thus constituting you know a complete answer at least within this four-fold framework that's been very helpful so that is where variational ecology comes in we're going to talk about it in the

context of building on
vne and what that is too so the paper is
variational ecology and the physics of
sentient systems
in physics of life review 2019.
ramstead constant bad and fristen
so the goal of the paper and again we're
going to return back to these things
so is specifically we review a framework
for modeling
complex adaptive systems for multi-scale
free energy bounding organism niche
dynamics
thereby integrating the modeling
strategies and heuristics of variational
neuroethology
vienne with a broader perspective on the
ecological nestedness of biotic systems
we extend vne beyond the action
perception loops of individual organisms
i.e active inference by appealing to the
variational approach to niche
construction v_a and c
to explain the dynamics of coupled
systems constituted by organisms and
their ecological niche
so the big question is how can we make
a multi-scale ecology framework that
builds upon
eco evo devo and also adds insights from
physics and mathematics and complexity
so first part of the abstract this paper
addresses the challenges faced by
multi-scale formulations of the
variational
or free energy approach to dynamics that
obtain for large-scale ensembles
we review a framework for modeling

complex adaptive control systems for
multi-skill free energy bounding
organism niche dynamics
thereby integrating the modeling
strategies and heuristics of variational
neuromythology
with a broader perspective on the
ecological nestedness of biotic
systems so they're as i said
stated in the second part here it's
going to be a framework which is theory
plus
tools that's going to be about modeling
complex adaptive systems as control
systems
what are they going to be controlling
something about the free energy
boundedness of their own interactions
with their niche
so rather than just thinking about how
the physiological or the neuro
aspects are related to constraints on
behavior
it's now also about behavior looking up
we extend the multi-skill variational
formulation beyond the action perception
loop of individual organisms by
appealing to the variational approach to
niche construction
to explain the dynamics of coupled
systems constituted by organisms and
their ecological niche
we suggest that the statistical
robustness of living systems is
inherited in part from their eco-niche
as niches help coordinate dynamical
patterns across larger spatial temporal
scales

we call this approach variational
neuroethology
we argue that when applied to cultural
animals such as humans
variational ecology enables us to
formulate not just the physics of
individual minds but also physics of
interacting minds across spatial
temporal scales
a physics of sentient systems that range
from cells to society
so briefly here's the roadmap section
zero
is the introduction they first talk
about
the variational or free energy
formulation
and specifically about how active
inference models and generative models
are related to it we're not going to go
through the figures in this
discussion though the second part of the
paper is about variational neuroethology
which is about how ensembles of markov
blankets underlying the organismal level
uh facilitate the
organism niche environment that was sort
of the neuroethology level of analysis
then they take that sort of organism
upwards
view of niche construction with
variational approach to niche
construction and the ontology of
affordances
and reach the variational ecology which
is really just connecting these two
at the organism and just making it a
multi-scale ecology and then subsuming

the behavioral component within it
all right so to
get to variational neuroethology
especially if this is um
new learning about active inference and
how it relates to
other areas in neurobiology some of the
comments about variational neuroethology
from
other um
authors in response to the articles and
just
points and comments that are really
sometimes apparent other times not
in the literature but areas for
improvement in theory one thing that's
cool about the active inference area is
it really draws people back to these
fundamental questions
that really were long term about top
down or bottom-up forces for example in
ecology or behavior
different uh the
you know role of condition of
historicity ergodicity
these are long time debates but now
coming into this new
area where we can at least address them
head-on so let's think about
that perspective and i'm trying to uh
think about how coming from a regular
behavioral sciences perspective
this might play so could this reflect a
fundamental distinction between the free
energy principle
as an explanatory principle for the
clearly bounded ergodic biological
systems and the greater complementary

forces at play that constrain complex adaptive systems in general including groups of organisms in their ecological niche should we restrict the fep to level the organism and explore how the model connects meaningfully to other key concepts and evolutionary systems theory about the ancient needs relation

so that's restating the question whether the fep

for some reason would stop applying at the level of the organism

and then it would break down at higher levels

the aim of this paper is that they're

going to address the challenges

these comments that people brought up about the vne

by addressing how it sits within broader level

questions and so there are

things that the field of behavioral sciences

has also been thinking about for example

you can watch one bird's behavior

it's what you're going to be tracking

but then it's going to be related to

everyone's behavior so at some point

it's going to be like studying

collective behavior

so what they're going to do is add the

variational neuroethology

which is the multi-scale plus the neuro

plus the behavior

plus this niche construction just to

additively create this variational

ecology

new acronym new paper why not
i thought um by putting together
ensemble dynamics or collective behavior
eco evo diva and niche construction
complex systems
free energy principle multi-scale
systems and active inference and
nestedness and embeddedness
and then kind of just think about how it
could apply to systems that
persist far from equilibrium within
this flow state valley of things that
are small and fast on the bottom left
to big and broad geospatially
on the top right but we won't go through
the figures in this
talk uh all right
to date vienne has this is before this
paper to date vne has only provided a
principled method of analyzing nested
and mutually constraining biological
systems and their complex adaptive
dynamics across spatial temporal
scales it is yet to offer a way to
individuate systems at scales beyond
that of the organism acting in its
environment
large-scale ensembles such as societies
or ecosystems that realize free energy
bounding dynamics
but of course part of the question is do
they and so
vne was really a theory that was about
the organism and the
ecosystem from the perspective and the
focus of the organism's dynamics in the
immediate environment for example
affordances

even if those affordances were subtle or
relied upon this skilled intentionality
or like you know learned behavior
or sophisticated counterfactuals all
these things that especially more
recently have been
receiving focus this did not um
articulate clearly
or at least had not been specified
specifically
that it was linked with a collective
behavioral or ecological or cultural
phenomena
which is uh as much a phenotype it's as
much biological as an organism level
trait like arm length or
genome composition there just isn't a
difference between
a bio cultural versus bio
in this variational framework it's just
different scales different mechanisms
but
it's not even about categorizing
different things it's about how you
model them and what they are um
so the variational neuroethology was a
more clearer
downwards projection about how things
like behavioral
and hormonal and neural time scales
developmental time scales
were related to organism behavior which
is physiology
neuroethology but the eco evo devo was
less clear
issue was cog gently articulated by
brian berg
and has spent kiramire in their critique

of variational neurology
they asked also whether the markov
blanket and b formalism leveraged by the
vne to individuate systems is adequate
for modeling phenomena at the scales
beyond that of a single organism
like socio-cultural dynamics since
phenomena at these scales may be too
transient or not sufficiently robust to
license the mb markov blanket formalism
which is defined in terms of conditional
independencies in weekly mixing
random dynamical systems which we'll
also
get back to but bhk
catalyzed this perspective and this is
gonna be broken up into like a few
different
sections markov blankets so
open parenthesis markov blankets are
these statistical using their words
conditional independencies
which is kind of like saying that it's
the opposite of the causal relationships
and so it's equivalent to finding the
causal relationships through their
violation of the independence so finding
which variables
are insulatable from each other in um
a sense if you if you think about what
kind of information
from a generative model you need to
understand which variables are
conditionally independent or
just sort of straightforwardly
correlated that's
often related to causal state estimation
about the world

markov blankets are unable to explain
slash

predict slash control ecosystems and
other collective behavioral

uh collective behaviors quote even
though it's collective behavior all the
way down

at scale because the markov blankets are
not robust enough to try
to capture transient or historical
phenomena

so to go to the paper they they only
work through one example

in the paper which was how this
hierarchically mechanistic mind
evolutionary systems theory can be used
within that vna framework to study the
scales of

human socio biocultural evolution in
humans

and there's this table provided with
some pretty simple slash

um just one line equations

about how different levels of analysis
would be modeled

but uh there was a critique which was
this downplays a lot of features of
ecology

and the formalisms that apply to
strongly

sensorily or strategically bounded
systems or certain spatial or temporal
time scales

uh is may only be weakly explanatory or
valid

at different time skills for a number of
reasons which is something that we're
going to return to

when we talk about the ergotic systems
so here's kind of a theory tree
the fep is on on the bottom it's sort of
the uh
the paradigm level foundation
because that's where we're gonna try to
make things emanate
from in two but again we could just
think about
what would be the effect of switching
that out too why not
from left to right we have active
inference tin bergen's four questions
evolutionary systems theory and then eco
evo devo
so variational neuroethology is like
tim bergen's four questions which was
what we heard the quote about
with an evolutionary perspective but a
simple
uh at least a sketch of one and then
more of an emphasis on active inference
and on
smaller scales of analysis the
variational approach to niche
construction vanc
was also drawing on the evolutionary
and the pluralist
tim bergen framework but drew a little
bit more from the eco evo devo and
document for inside like
just it has that orange one
and where ve comes into play
is with bringing in this synergetics
idea
from the variational approach
and also the collective behavioral
approach and then just

again making a new acronym kind of
more specifically combining previous
acronyms

i think it it uh it's worth unpacking
it's worth tracing out some of these
relationships

and just seeing what parts get subsumed
into what

but ve is just um

it's just a more high resolution area
where mapping can now be performed
across these different
domains so it's kind of like a mapping
domain

so here's some of the specific parts
where they talk about that kind of a
mapping domain

in this multiscale framework active
inference is inherently a group activity
that is the entire ensemble of nested
markov blankets are bound
enslaved and constrained by dynamics at
higher scales while the lower
microscopic scales furnish the
macroscopic

states at any given level this
construction evinces exactly the same
circular causality that underlies
synergetics hakan not fuller

however here it is generalized to a
recursive hierarchy of scales

i.e the hierarchical composition of
blankets of blankets

intuitively the dynamics at one scale
provide constraints technically they
establish probability gradients on
dynamics at other scales

active inference destroys free energy

gradients at each scale
under the guidance or control of a
generative model at the scale above
this guidance is exerted through
influences on sensory states where
circular causality
means that the action of any markup
blanket in an ensemble of markov
blankets could be involved in sensing
action or perception depending upon its
role at the superordinate scale
i.e it has a sensory active or internal
state at the
level above so there's this kind of base
all the way up and down
uh approach that they're proposing that
from the level of the organism can go
downwards into the cellular the neuro
and all this level
and through time through the
evolutionary through the tin bergen
quadrants
and then also outwards through niche
construction
in the multi-skill setting an effective
state at any scale
like let's just think of our temperature
becomes the expression of an eigenmode
of blanket states
namely the principal eigenvectors of the
jacobian the rate of change of flow with
respect to state
these mixtures are formerly identical to
order parameters and synergetics that
reflect the amplitude of slow or
unstable stable eigen modes
in terms of center manifold theory they
correspond to solutions on the slow

unstable or center of manifold
in short the markov blanket of a system
or particle at any scale
constitutes an ensemble whose ordered
parameter sub 10 blanket or internal
states the scale above
so that's like the temperature being
above equilibrium with the room
is because it's part of a sort of closed
or semi-closed regulatory system that's
keeping
molecules that are in one part of the
room constituting my body
a different temperature than the rest of
the room
note that the constituent microscopic
states of an ensemble are always blanket
states
although their order parameters could be
blanket states or internal states at the
microscopic scale above
so a neuron could be sheathing something
from a temperature perspective from one
side but receiving information in a
different
uh conceptualization of the blanket
network this follows from the fact that
only
the only states that matter are those
that influence other blanket states
effectively all we are doing here is
applying the enslaving principle or
center manifold theorem recursively to
mark out blankets and mark on blankets
imagine you are an employee at an
institution where you transact your
microscopic affairs with other personnel
to self-evidence your prior belief that

you are good at your job
this would entail responding to
corporate or institutional goals that
emerge collectively i.e an implicit
generative model at the macroscopic
level
note that your job may be homologous to
an internal state at the institutional
macroscopic level relating only to other
employees alternatively you could be
working on reception
i.e a sensory state or issuing press
releases ie in active state
and this is actually an image from a
paper that
team com wrote on 99 2020
and one of the figures and it was about
how
remote teams and a lot of other kinds of
groups that might not be traditionally
considered remote teams can be
explicitly mapped and from systems
engineering perspective modeled
as this nested markov blanket framework
and so you have for example the team
members who have their own digital
interfaces their own sensory and active
states digitally
some of these team members can be
augmented or can be bots themselves
and they're interacting with certain
databases and then they're able to also
undertake
team operations this is where the
variational approach to niche
construction
comes in the anc exploits the expression
of expected free energy as expected cost

plus ambiguity to propose that agents
upload as it were
much of the leg work in computing
expected free energy to their
self-tailored constructed environment
more precisely it argues that niche
construction can be cast
as the collective activity by which
organisms act on their material
environment
to create unambiguous structure which
can then be leveraged via active
inference
like creating books and then people can
scan their eyes over
lasting changes to the niche capture the
fact that environmental cues function as
an unambiguous indicator of the
affordance of action possibilities
like if you're culturally sensitized to
know what a book looks like
you know what it's for you know that if
you scan through the first three pages
with your eyes
and it's culturally specific whether the
first pages are going to be on this side
of the book or on this side of the book
or
whether the text will read this way or
this way
this structure can be cast in terms of
epistemic resources that flag the
actions that resolve the ambiguity
associated with future observations
while conforming to the prior
preferences of the organism entailed by
its generative model
so you're in the library you're doing

some optimal foraging
for a book about a certain topic and you
expect that there's a certain structural
organization to the books
and that if you find something where
it's written on the back of the book
that you think it's the book it is
you expect that the title page is going
to conform with that so these are like
epistemic resources that are more
visible
more scannable that's what allows that
information to be
found and therefore leveraged so that's
niche construction
in terms of direct epistemic resources
but you can also think about
in cases like social insects where their
extended
niche their nest is something that they
pass on potentially even
intergenerationally and that the
structure and the form of the function
are all
linked to their collective behavioral
algorithms with the architecture
influencing their behavioral
proclivities
and there's some interesting work by
professor pincer wolman
on that in the heart of strands
then that physical structure becomes a
epistemic resource as well in that it
becomes part of their extended cognition
all right so
for this last section of section
07 of the live stream my friend sean
had a great question and so for this

last section it will be a little bit
about
ergotic systems and as with
you know all other areas of this sort of
discussion
it's uh it's a playful and it's a
partial recap of the ideas so
if someone knows more i would love to go
deeper into some of these
ideas i just kind of laid it out in a
way that hopefully
might relate to uh an informative way to
onboard onto thinking about things from
an active inference perspective
and there's definitely perspectives on
youtube and elsewhere that are a lot
more rigorous and comprehensive than
what i'm about to go through
but this is just gonna be a couple of
minutes on
linking ecology which is what we're
trying to explain
we're not just trying to make theory
after theory we really want to return
to the ecology and make variational
ecology a theory about ecology
that allows us to do the things that we
want to do in ecology whether that's
conservation or just understanding
so many other things that ecology can
teach us but we're also
going to talk about erodicity about cog
ergo sum and of course active inference
all right so what sean asked
was are thinking beings or their
environments
weakly mixing ergotic systems if not do
they fulfill

some of the assumptions or axioms of the
mathematics underlying the fep
so this was kind of a great question by
sean because
it really came at a perfect time when i
was preparing
these slides and if we go back to the
critique slide where the very own
variational neurology was being
presented
remember it was that there was the
critique of the vne
that got us to the v the response to
that that got us to the ve
as far as connecting the organism down
with the vne with the organism up with a
niche construction just connecting the
two points into a line
and calling that variation oncology and
the the the critique was that there was
going to be some violation
of some of the lower level sensory um
coherentness of say an organism
and that because of that it wouldn't
that the same formalisms wouldn't apply
and so this is someone coming from a
more like thermo statistics
um background and seeing a very similar
phrasing a different way and just asking
a really interesting related question
and as he put it the argument of the
free energy principle is two parts
the first calls on the lawful dynamics
of any weakly mixing
aka organic system that persists in a
changing environment
from a single cell organism to a human
brain these lawful dynamics suggest that

internal states can be interpreted as
modeling or predicting external causes
of sensory perturbations
aka embodying and acting a generative
model of adaptive policies and
phenotypes in other words
if a system exists its internal states
encode probabilistic beliefs about
external states
and so when i read that you know i
didn't know it's like is he
is he summing that up as like the
insight as like the final
distillation of something that he finds
deeply true or is he bringing this up as
sort of a well that's a tautology that's
a circular
logic or that's an argument that holds
nowhere
so i i didn't know but uh it turns out
that
there is a paper from 2015 called i am
therefore i think
so um this last little part is going to
be about that paper
which i hope will be interesting whether
you're familiar with these ideas or not
but also with some of the uh
mathematics behind the ergodicity and
about why there's still
significant reason to understand how
ergotic systems non-ergotic systems are
related to the fep
so i am therefore i think is uh kind of
related to this i think therefore i am
calgary or some but
of course it's just a meme so here's a
few other versions

i think therefore i am not here or
a comic where it says i think therefore
i am and then the other person says i
don't therefore i am not
and uh also the cog cogito ergosome
i think therefore i am i think therefore
i overthink either i think therefore i
regret i think therefore someday i won't
so all these different implications what
what you know what are the real
implications what logic
can we can we make claims about well
that's not exactly what this is about
this is going to be about two kinds of
systems
which are ergotic systems or not or
or ergotic systems and this isn't
something that's um a hyper abstract but
it definitely
does recall a lot of generalizable
principles and math
so to make it very specific and
empirical
we're going to discuss it in terms of
the ergodic hypothesis
so it's a specific hypothesis that we're
talking about
that means it's something we're going to
test with data and the hypothesis
is we hypothesize that mixing through
space
is like mixing through time and let's
talk a little bit more about that
because it's kind of a
wild and interesting idea so the first
thing to know is that it's a
hypothesis this ergotic hypothesis it's
not the ergotic

law or the ergotic pattern or uh
anything like that it's hypothesis
that's uh
can be uh tested using data and
statistical models
that are transparent and reproducible
and just
open to understanding how changes in
their parameters influence the
kind of conclusions that are reached so
that's something
about what we're discussing here with
the ergotic systems
also the ergotic hypothesis depends on
the spatial and temporal scale
and then also the measurement or the
analysis space and
potentially this comes into play a lot
more when you're really getting down
into the weeds of the
math or the application but often the
way that you measure something
or which sections of the time series you
allow for burn in for example
or mixing these things are all really
critical and what
what scale whether you're looking at a
tiny tiny tiny
cube and seeing if there's random
association between chemical particles
for example versus looking at a
one meter cubed uh blocking of the world
so the coarse graining matters a lot now
ecology and history are often strongly
critical of applications of
ergotic methods or that methods that
often
uh are used to model ergotic systems so

first let's
talk about the ergodic hypothesis and
what it actually says
and does and then talk about some
ecological critique of it
and then see how the cog ego sum comes
into play
all right so the ergodic hypothesis is
defined
by the time average being
equal to the space average so
that is uh again this sort of mixing
through space is like mixing through
time
idea and one way to think about it is
like
if you have a box with a gas in it
and you're going to trace one molecule
and take snapshots of it at 100 times
would its trajectory in some statistical
way
be similar to if you just selected 100
particles in the box at the same time
or if you let loose one molecule bounce
a hundred
times by itself all these different ways
of asking it
but it kind of relates like is it
basically fair
to sample you know one roulette table
through time or a hundred roulette
tables at the same time
every roulette spin is like the same so
in in
it's very different than um uh something
where
the uh strategies evolve through time
for example you might already even be

seeing

and one more slide of just the definition is that there's this generation of an ensemble so that's sort of

already for saging this ensemble group selection dynamic versus the time element and this paper was very interesting

that i found while preparing for this stream and i'll just go through it because i think it's also a really cool story about a philosophy of science and about

how the way that we think about different hypotheses and axioms and principles

really influences the kind of science that we do so it's called from the ergodic hypothesis and physics to the ergodic

axiom in economics by kirsten 2015. so

the the work is summarized as this contribution seeks to clarify the specific relation between the idea of non-ergodicity which is drawn from statistical mechanics and its role in and for economics and finance the identified change of status is how the ergodic hypothesis in physics became the ergodic axiom in economics therefore we follow the idea from rational mechanics of gibbs via wilson to samuelson and eventually from the latter into rational economics the methodological spillover from the natural sciences to economics and the

assumption of ergodicity in particular
enabled and shaped the mathematicization
of economics
seen since the 1940s so it shapes the
fields direction
depending on how people use methods from
ergodic uh
assumptions or not and this was a nice
diagram where
boltzmann and poincarey and gibbs
and how it transduces through different
generations and it's kind of like a
kind of like a citation network here
and i think a great example which was
used in the paper
is this st petersburg paradox and this
is a really fun
kind of game theory and thought
experiment complexity
uh classic that's related to economics
so i'll just read it as it is from the
paper in seventh it's called the saint
petersburg paradox
in 1713 swift mathematician nicolas a
bernoulli
posed in an exchange of letters with a
french mathematician pierre montmore
montmore
the following problem what does a
reasonable price
p for a ticket for the following lottery
so on the first flip it's gonna be a
fair
coin and uh if you win
you can either take one dollar and then
uh game's over or you can play again
and then on uh
the second flip if you keep playing so

there's

there's multiple versions in this

version in here

it either gives you a dollar or you keep playing and so you get a payoff at each time

um if uh another version is where you can decide to keep playing and then you can imagine that getting a 50

chance of that first one is kind of like

now the games are very similar

because you can either um either by

by force or buy not so in either case

you can ask yourself how much would you

pay to play either of those two little

variants of the game

and um it turns out that people don't

want to pay infinite money to buy in

so especially the one where you either continue or get the payout

the expected value of of all the

ensemble of trajectories one in a

million is going to make you know two to the n

money and so one in a million is going to make so much money

uh it's the expected value across the

ensemble with the geometric mean

or the arithmetic mean would be like

very high but it still wouldn't be worth that much

why so let's go into that why does that

st petersburg petersburg

paradox uh challenge ergodicity

one answer is it's about pragmatics

your life is not ergotic you're not well

mixing across

um you know different places to eat
across different places to sleep
there's a regularity to your life and so
especially
considering it in the sort of niche
construction for your own
uncertainty reduction slash precision
maximization
even putting that all aside it's just
your own
singular trajectory from a strategic
perspective
doesn't resemble the assumptions
that are under the
distribution-conforming norms of
well-mixed and gas molecules that's for
sure
and another way to say that is that the
realized risk profile through time
is not the same as through space so if
you're playing that game you know who
wants to be a millionaire and it's going
up okay
i've got 12 coin flips in a row i'm at a
thousand dollars or something like that
i could take two thousand dollars or i
can go home with nothing
at some point it's just not going to
make sense
to go double or nothing on some insane
value because there's this realized
trade-off between risk so that's not
and that's part of this historiosity and
non-eradicity of your own trajectory
ecosystems may not be ergotic so that
would be like
because of secession dynamics or
non-linear

perturbations or resilience properties
ecosystems
may or may not have this sort of
roulette wheel property
or other nice
aspects of uh high or low dimensional
systems that physics
prefers so there's interesting
theoretical work oh well imagine if
every patch the ecosystem was like a
spin glass that would be a phase
transition
but in the real world it's always a
little bit messier than that
so then just take that sort of st
petersburg paradox and then
literally the money one now adding niche
construction and so i was just thinking
well
yeah what if what if every city in the
country gets to play this paradox
and then one city or you know one
individual
wins just a ridiculous amount and a few
people with a small amount
most people with nothing and then that
city does infrastructure investment
so how does that change the way that
future people would be able to
for example play the paradox later
so that's niche construction so now
think about the historicity and the
niche construction whether that's social
insects or humans
and you can see how some of these
ergodic assumptions
as they apply to cultural
behavioral systems may be like

a little bit tenuous or metaphorical at best

here's a few nice figures from this paper though too

so on the bottom right there's this ergodic case and so one of the features of ergodicity is this time reversal like the molecule bouncing around the room it's kind of this

general time reversible family of models that allows extremely

um interesting bi-directional

computation in a way that historical systems don't

and so timelessness exists in a current t equals zero

that's the present moment and then

there's like this uh analytical time

like reaction coordinate for a chemical reaction when the two molecules come together

it's not graphed in terms of time but reactions do have time distributions that they occur on

but the way that that is represented is on actually a free energy landscape gibbs free energy

with um a reaction coordinate because that's like this ergodic

hypothesis about the way that these reactions are reversible through time

in contrast there's the non-ergodic case on the top

and in the top there's a symmetry break in the present moment

so that means that there's events in the present moment that change the distribution going forward

in a way that doesn't make time
reversibility a tenable proposition
and that's because they happen in
historical time
so that's kind of a cool idea and then
there was yet another great figure from
this paper
where with kind of a lot going on but i
think it's worth looking at because it
was so cool
so first there's the not analyzed not
analyzed is just before your data window
is available
then there's this realm where you use
the ergodic methods ergodicity it's
almost like the name of a city
and what you see in this window is the
spurious quote calls a relationship in
your observation window so let's just
say like you know between
bitcoin and and exports of some other
resource or something
and even if you everyone knows
correlation isn't causation but then
there's methods like dynamic causal
modeling
ranger causality and machine learning
methods that
try to quote learn so there's so much uh
in the way that people use methods
whether or not they think they're
they at the very least will say they're
using data from the past to reduce their
uncertainty about the future
which is what led to financial collapse
it's what uh
it's what leads to a lot of strategic
strengths and weaknesses

in a lot of different areas so that's
the ergodic
fallacy valley now
the non-causal window is like let's just
say where
there's a different window of time
whether it's
window two or window four and under
either of those time intervals under
consideration
the same factors that were at play and
one may not matter that much like if you
observe the first
post forest fire three weeks the rules
that dictate that regime
let's just say the spurious causal
relationship oh nitrogen is a limiting
factor or
oh insects don't play a major role in
this process
that may or may not be true in a future
time window that's not overlapping
or partially overlapping seems pretty
fair pretty valid i think it's a great
figure and then also there's like second
and future symmetry breakings as well so
even if you go well
there was a major systems change and the
present moment or
at x time but uh
things are still the same in this regard
this invariant still holds or this
measure is still
valid but then they they can't
guarantee that there's going to be this
uh invariance in a future arbitrary
symmetry break so what are the
frameworks that will carry us through

these symmetry breaks
and not just have very weak underbellies
with respect
to things that we don't even perceive on
the horizon
so let's return to debra's
paper about fusion of behavioral ecology
and ecology and um
talk about whether simulations
variability historicity and
other sort of not quantitative questions
but qualitative ideas
it challenged this ergodic hypothesis so
she wrote in behavioral behavior in the
illusion of ergodicity and ecology
it's a section this paper ecology often
represents ecological processes in terms
of the functional relation among
average values for example to ask
whether an invasion will be more
successful
when there are many species in the
community to ask how an average
propaganda will fare when the average
population size of many local species
are high
methods for evaluating ecological
outcomes often assume ergodicity
that all samples are homogeneous and
that populations being sampled are
invariant
this can be misleading behavior is a
response to conditions
but individuals vary in response and
conditions change
the result is that ecological processes
are not ergodic
samples are not homogeneous and

processes can change over time
very nice in a well titled section
and to kind of add on to this
perspective and
return to this question about
non-ergodicity in historical processes
like ecology
here's a paper in 2016
replaying life's tape simulation
metaphors and historiosity in steven j
gould's view life
so not a endorsement or critique of
steven j Gould it's to bring it up as a
philosophy of science point just as any
other
paper i would address so this is from
the paper and it is a 1974
paper of ralph and Gould and what they
did is they
did simulations with where branches
uh node branching rates were a parameter
and then also there was like mutational
changes and they're like wow the
extinction
distribution looks like
we see in the fossil record and the
morpho distribution
looks like we see in the fossil record
maybe
selection doesn't matter or maybe it's
just random stuff
and to kind of build on that in another
figure in the paper
this is two outcomes of the simulation
that they later ran so
this is probably a later paper from from
them or from from colleagues
that use again kind of like a birth

death

process so like the lateral line is a

birth of like a new clade

speciation event with a uh

of some type maybe with niche

displacement maybe not

and then there's also the population

size growing or shrinking by a small

amount but when it reaches

zero it kind of goes extinct and so they

were

seeing wow we're seeing big variation

between quote clades or

different levels of organization in

terms of their biodiversity or in terms

of their species number

or their morphological diversity so

again maybe it's just the random

punctuated

equilibrium events that end up driving

the ship

now this comes down uh and you can come

down on the stephen j Gould's side or

not

on this question whether the

historiosity

of the past the uniqueness of the

trajectory or the

the uh uh i guess the timeline that we

appear to be in whatever your feelings

on

the his on the historiosity of that the

question is really from an

instrumentalist point of view from a

functional point of view

can you use these methods that may have

statistical assumptions

underlying them that assume ergotic

processes

so for example when you do a t-test it's like making the assumption that with respect to other variables all else is equal but

and everyone says oh well of course if you if you sample group a and country one group b and country two

it's conflated but the t-test doesn't know that

and so where we don't know that the t-test doesn't know that

and more complicated machine learning methods

are only able to do this in a really more nuanced way

in certain situations especially with large data sets so

for the stephen j Gould camp it was the incidental long-term deep time events black swan events that drove the ship whereas another way to think about it in the one that i personally would probably end up emphasizing at the end of the day is that it's actually selection on the morphology locally over that deep ecological time

there's also the day-to-day of those little trilobite creatures

having to live in their niche and so selection is continually on their morphology and behavior

and then yes events happen over longer time periods and the whole question about multi-scale biology is how to integrate those two

rather than saying well oh well clade

formation is random versus oh it's
really strongly driven by niche
partitioning
so to kind of actually close this with a
with a
cogeto ergo sum uh part
here's some figures for me i think or i
am therefore i think paper
with fristen and mathias so
this left side is i am ergotic
therefore i think let's just you know
see what the authors say
because with all these uh critiques from
ecology
and from cultural biologists on
ergodicity
and on the brain the mind and everything
so here is
sort of a two-stroke engine and it goes
from rez
extensa which is sort of the the the
behavioral action policy selection
flowing out
and then the res cockpit is like the
belief partition of the brain
so the belief could be i believe that
the training is going to come in five
minutes
and that would be like corresponding to
a detachable thought or
debate whether it's a representation or
whether it's a metaphysic uh component
and then this is just one partitioning
they're thrown out there we'll see
another one in a second
and then this free energy functional is
being used to select the maximum
likelihood policy and so that's why it's

doing
um this pursuit of the
minimally surprising across the
variational scales of analysis of
surprise
is trying to select the control policy
that's going to bring
sensory observations into alignment with
a deep generative model
so the belief i am ergodic could lead to
behavior that at least pursues
ergodicity
is that enough and then
uh they kind of build it out one another
level which is i think i am ergodic
therefore i am so that kind of is
an inversion of i am therefore i think
i don't know how you get from here to
there maybe there's other figures to
make or other arrows to draw but it is
interesting to think about how basically
it's a little bit unpacking the
um uh res cognitives beliefs
from a bayesian perspective into the
posterior belief
and then the likelihood and prior belief
so that's like the
um now there's like a bayesian
two-stroke engine up there
and then out of the free energy
functional reflecting the likelihood and
bayesian
beliefs is also um
the policy that sort of relates to our
discussions on the
uh generative model being a policy model
all right here's kind of a funny figure
with uh focusing on the ear of this

statue

and just uh

uh it has equations on

the ocular motor being linked through

the action so action causes the ocular

motor to move

and the eye sends back visual sensations

and proprioception

so that's sq and sp and it's like that's

the interface for the eye that's the api

for the eye that's what we have access

to

and that's what we can model and it's

that dual instrumentalism the

one layer is like yeah these are the

areas of signal that the eye gets and

receives

so from an engineering perspective the

scientists instrumentalism is like yeah

it's good enough for us just to model

the system that way

because there's basically nothing that

goes outside of that flow state even

though each one is complicated

but then the second one would be this is

actually how the eye works and it turns

out that some of these hypotheses

about how the actual streams of

information or communication should be

in the brain

that they are concurrent with some of

these inference schemes

which is interesting and this figure

was about how like looking at the same

statue

there's the stimulus which is the actual

image itself

and what the visual input is is that is

the retinal cells which are activated
let's say by
edges and then the salience could be in
terms of a
deep generative model as far as what
area of the sample would be or what area
of the
visual field would be most optimal to
sample ergo
which action policy would be optimal to
undertake and we see this
sampling which is a um
a little bit outlining the face if you
kind of look at it overlay it
because these are the optimal places to
forage
so to speak for information to resolve
what it is
so um i'm just going to go through this
last
figure from i am therefore i think
because i think it's kind of interesting
and
funny so here's how we can walk through
this
a behavioral sequence is embodied so
that's the ocular motor
red dots moving around it's like
so the behavioral sequence is embodied
or enacted and also encultured and
embedded that's the whole
variational approach so right now we're
focused on the ocular motor
dimension to really unpack it but
it's something that we could have moving
up like in one culture
they learned that avoiding eye contact
is

you know or people's status is signaled
on this part of their body
or no it's in the eyes where you
determine something
so this is not just a
a one-size-fits-all solution for human
vision this is about a model framework
that we could use to model what we want
to model but
a behavioral sequence is embodied by
acting
to converge towards beliefs about hidden
ocular motor states so right there on
the y-axis is this hidden ocular motor
state
and then there's the proprioceptive
expectations
so that's like if your muscle with one
side like if your bicep contracts
your tricep will relax and if that
breaks down
if that communication of expectations of
oh hey i'm contracting or i'm contracted
if that breaks down you get a lot of
different types of muscle
phenomena ranging from you know pulled
muscles and strains to
um like post injury or even without
injury
changes in the muscles perceived healthy
range of motion or
um limit or minimum maximum force and
that has to do with a lot of the
proprioceptive information being
integrated with the beliefs about the
hidden states
what that results in is that observed
sensory states

um which uh uh these are the visual samples
so that's we're taking one the hidden ocular motor states
we converge towards belief on those what that results in
is the observed sensory state so here's like an i an i an
i and you know there's five dots here and so you're going to get more sampling on the eye but that's kind of what the camera
sees and then it says absolute data or surprise or prediction error because this here is like showing the visual samples as kind of camera snapshots of what is being seen
but in the predictive processing framework you could imagine that actually what's coming back is like a parameter update message passing thing that's actually about deep expectations about
what to see so that's not even included in this model but that's part of other models
and what that does when the sampling is like oh i expected an eye there i looked at her uh you know third eye
and i didn't see it visually you know hypothesis confirmed
so that type of sampling is also informative
and so what we see on this graph is that the expectations
about hidden perceptual states so are sort of fuzziness on the whole map
uh decreases through time

and what that results in is that the uh
corresponding percept remains sharp and
concurrent with a coherent generative
model
through time so i always just think
about vision in this case
like your whole visual field unless
there's other secondary
issues is clear and has color vision
across the entire
field and so this is maintained
even though there's a large blind spot
your resolution is lower
outside of your center area and also
there's not as much color or perception
outside
um there's a bunch of other visual
illusions too so it's clearly we're
perceiving our generative model visually
and why not in other areas too when it's
as clear or clearer so we have this face
just coming
into clarity based upon a deep
generative
prior which is why when there's things
that you don't know how to extract the
information from
it's hard to know how to do ocular motor
scanning let alone
how to do the semantic mapping now
isn't that the big question is will this
kind of
static face fixed image scanning
to resolve uncertainty will that level
of functional closure relate
into eco evo scales of analysis
so you could imagine this is just a
static face doing classification but

would this extend into adversarial
relationships with one person
would this extend into deep
counterfactuals with valence
and skills and
team reformation hypotheses all these
things playing out that
are happening in social scenarios would
that be able to play out this is just
eye scanning so
will that uh level of analysis
of the group or the community the
evolutionary trajectory the cultural
unit
the ecosystem any ecosystem
will that level of analysis lend itself
to this multi-level
uh framework and isn't that
the question so that is all
i thought i would close with it as an
opening to
whether people thought yes or no or
i think i'd like to know this piece of
information
before we made a decision i'm i'm open
you know i want to learn more
i want to figure out what that actual
answer is
is it going to be that we'll have some
very tight
and coherent markov blankets that uh we
can just do simulations and understand
the interactions between
and then let that higher level parameter
just move freely
or will it really be the case that there
are order parameters
that are just beyond what we can

currently
understand as mapping onto this
framework or some other framework
who knows um anyways that was kind of an
interesting discussion so i hope you
liked it and i just say thanks for
participating
we will provide a follow-up form to live
participants
feedback suggestions and questions
always welcome
and yeah just stay in communication
we're gonna have
number 7.1 and 7.2
in the coming two weeks and uh if you
want to participate
in your listening before them then get
in touch
also we'll hopefully still be
looking for papers to read and for
people to have on discussions
as long as you're listening to this so
just give
us a comment or a message and it'd be
good to get in touch
so any thoughts especially on this video
on
where does the fep fit into all this
where does active inference fit into all
this
what are we looking to understand with
variational ecology how are we going to
use
variational ecology how is it going to
map onto classical ecology
what ecological questions are we going
to be able to address first
how is this going to relate to the kinds

of explanations
and design solutions that ecologists
have worked with before
versus propose deeply uh
hypothetical systems a lot of
interesting questions plus of course
hearing from everyone else on their
perspective on the implications or
strengths or weaknesses so anyways cool
thanks for listening i hope that you had
a good
time and found some parts of it
interesting
and i will talk to you later